
A Decoupling Algorithm Based on PID Neural Network for Multi-Channel Active Noise Control of Nonstationary Noise

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It is critical to study multi-channel active noise control (ANC) systems to satisfy the requirements of noise reduction in multi-target positions. However, the complexity of the system may increase as the result of an increased number of sound channels. In addition, multi-channel coupling affects the stability of a system. In this paper, a decoupling algorithm based on the Proportion Integration Differentiation (PID) neural network and the filtered-x least-mean-square (FxLMS) for the multi-channel ANC of nonstationary noise is proposed. Due to the nonlinear characteristics of the PID neural network, the coupling problem can be solved through the algorithm. The performance of the novel proposed algorithm is verified by comparing the simulation results with the results from the traditional FxLMS algorithm and the FxLMS algorithm with matrix decoupling. The results illustrate that the convergence speed of the traditional FxLMS algorithm is similar to that of the FxLMS algorithm with matrix decoupling, while the proposed algorithm converges significantly faster than the other two algorithms. In terms of control performance, the proposed algorithm executes the best and the residual error signal has the minimum amplitude, followed by the FxLMS algorithm with matrix decoupling and the traditional FxLMS algorithm. With the advantages in convergence speed and control performance, the proposed decoupling algorithm could be suitable for multi-channel nonstationary ANC.

1. INTRODUCTION

An active control algorithm is the key to an active noise control (ANC) system. Essentially the performance of the entire control system is determined by its quality. The earliest active control algorithm named the least-mean-square (LMS) algorithm¹ is based on the stochastic gradient method, and is characterized by simple and effective recursive calculations to find the best weight vector. However, a time delay problem is caused by ignoring the physical channel between the secondary speaker and the error sensor. To overcome the adverse effects generated by the secondary channel, Burgess² proposed the filtered-x least-mean-square (FxLMS) algorithm. In this algorithm, the secondary channel transfer function is embedded as a filtering link into the channel before the reference signal is input into the controller. The FxLMS algorithm has become the benchmark algorithm for the ANC system due to its clear physical mechanism, low computational complexity and simple implementation. Nevertheless, the convergence of the adaptive process in the FxLMS algorithm is relatively slow if the autocorrelation matrix of the reference signal has a large range of feature rooted dispersion. Different improvements have been proposed to solve the aforementioned problems, e.g., the normalization,³ the variable-step-size,⁴⁻⁷ the logic,⁸ internal model⁹ and blocks⁵ FxLMS algorithms. The main purpose of algorithm improvement is to facilitate its hardware implementation, speed up the convergence rate, and reduce the adaptive steady-state error.

The research on active control technology mainly focuses on the single-channel ANC system that is suitable for active

noise reduction in the designated area. Nevertheless, it is difficult to satisfy the active noise reduction requirements of the multi-target area. In recent years, considerable attention has been devoted to the design of the multi-channel ANC system for improving noise reduction performance. The multi-channel ANC algorithm that is based on the single-channel FxLMS algorithm and extends the number of secondary actuators and error sensors has become the most commonly used algorithm in current research. However, this algorithm has many limitations. First, as the number of channels increase, the computational complexity of the algorithm increases dramatically. For the traditional multi-channel FxLMS algorithm, if the number of reference sensors, secondary actuators and error sensors are denoted as N , the computation complexity is proportional to N .⁴ Thus, as the number of channels increases, the calculation scale of the control system is enormous. Second, as the number of channels increases, the coupling among channels deteriorates the stability and the control reliability of the system. The failure of the whole system can be caused by the instability of any single channel. In order to overcome the above aforementioned problems, researchers have made significant progress on the multi-channel FxLMS algorithm. A novel nonlinear adaptive algorithm named the diagonal structure bilinear FxLMS (DBFxLMS) for the multi-channel ANC system was proposed by Chen.¹⁰ Simulation results showed that the algorithm had better control performance than the second order Volterra FxLMS (VFxLMS) algorithm. Chung¹¹ presented a normalization multi-channel FxLMS algorithm that can be applied to adaptive filters for multi-channel active noise control. Simulation results showed that the pro-

posed algorithm presents better convergence characteristics under non-stationary environments. A new formulation was presented by Mazur¹² to introduce the distributed version of the Switched-error FxLMS algorithm together with adaptation rate enhancement. A new architecture based on multi parallel folded branches (MPFB) was proposed by Shi¹³ in order to reduce the computational load of the feed-forward multi-channel FxLMS (FFMCFxLMS) algorithm. The branches were parallelized, and multipliers and adders were reused in each folded branch. Thus, the trade-off between throughput and hardware resource utilization was balanced, and the computational complexity was reduced.

Although most of the research the complexity and stability of the algorithm are studied on multi-channel control systems, the channel coupling problem needs further study. The stability of the control system is affected by the mutual interference coupling¹⁴ between the secondary channels. Furthermore, any error signals must be used for updating all control signals, which leads to an increase in computational complexity. A matrix decoupling optimization algorithm based on feedforward compensation was applied to the multi-channel ANC system by Weipeng Gao.¹⁵ The decoupling optimization matrix was inserted before the secondary channel matrix to realize a one-to-one correspondence between the control and error signals. Consequently, the control system became equivalent to multiple single loops, which improved the system stability and convergence speed. Matrix decoupling relies heavily on the precise mathematical model of the controlled object.^{16–18} As the actual noise control system is a nonlinear and time-varying complex multivariable system, the matrix decoupling is suitable for a few channels and cannot meet the actual control requirements. In the case of the multi-channel control systems, the calculation of the decoupling matrix is more complicated, and the control results thus far, cannot be improved.¹⁹ Therefore, the method for decoupling the multi-channel ANC system should be further studied.

The development of an adaptive intelligent algorithm is a new trend in active control. The combination of neural network algorithm,^{20,21} fuzzy algorithm²² and adaptive algorithm²³ opens up a new path for active control of vibration and noise. Kurczyk²² proposed a comprehensive method using fuzzy control to replace the inverse modeling of magnetorheological dampers, which expanded the application conditions of the control system. Based on the above theory, dealing with system instability caused by mutual coupling among channels, a novel multi-channel ANC method based on the combination of the FxLMS algorithm and Proportion Integration Differentiation (PID) neural network decoupling control algorithm is proposed. The neural network^{24,25} is an active network that uses neurons as nodes and utilizes a certain network topology to describe almost any arbitrary nonlinear relationship. The PID neural network control method that has been used for a variety of control systems, such as the heating, transtibial prosthetic limb,²⁶ the reactor core control system,²⁷ mine main fan switchover system,²⁸ and the hydraulic quadruped robot,²⁹ has a simple structure, stable working state, wide application, and the advantages of non-linear information processing. The results show that this method has good dynamic quality, robustness and anti-interference ability, and can effectively increase

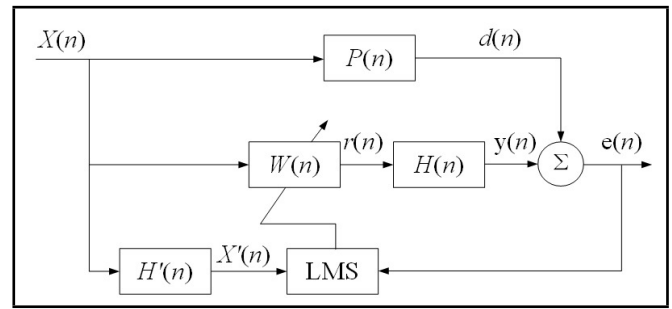


Figure 1. Block diagram of the FxLMS feedforward active control algorithm.

the control speed and improve the control precision.

In this paper, an algorithm combining a PID neural network decoupling algorithm and the FxLMS algorithm is proposed to solve the channels coupling problem. The coupling among the channels is regarded as the channel disturbance problem. Based on the traditional FxLMS algorithm, the PID neural network is used to adjust the control parameters of the ANC system to obtain optimal control. Additionally, the decoupling and control problems of the multi-channel active control system are processed in parallel. Normally, the noise is nonstationary noise that changes with time. Aiming at nonstationary noise, the proposed method uses the sum of sinusoidal signals as the input signal to simulate. The simulation results are compared with the results from the traditional FxLMS and the matrix decoupling FxLMS proposed in exiting literature. The results show that the novel method has fast convergence speed and good control performance.

2. MULTI-CHANNEL ACTIVE NOISE CONTROL ALGORITHM

2.1. Multi-Channel FxLMS Algorithm

The control goal of the ANC system is to reduce noise in a certain three-dimensional space. Considering the complex and changeable noise field environment in a cockpit, the multi-channel active control scheme should be adopted. The core of the active control system is the active control algorithm, and the principle of the algorithm is explained in detail. According to the structure of the feedforward active control system, if there are I reference sensors, J secondary sound sources and K error sensors in the control system, the numbers of primary and secondary channels are equal to $I \times J$ and $J \times K$ in the whole system, respectively. The filter length and primary path are L and $P(n)$, respectively. Figure 1. gives the block diagram of the FxLMS feed-forward active control algorithm. $X(n)$ is the input signal, $H(n)$ is the secondary path transfer function, and $H'(n)$ is the estimated value of $H(n)$ obtained through the identification of the secondary path channel. $X'(n)$ is the reference signal matrix, which is obtained by the convolution of the input signal $X(n)$ with the estimated value $H'(n)$ of the stimulus channel.

Suppose that the reference signal vector received by I reference sensors $x(n)$ is as follows:

$$x(n) = [x_1^T(n), x_2^T(n), \dots, x_I^T(n)]^T; \quad (1)$$

where the i -th reference signal vector $x_i(n)$ is represented as:

$$x_i(n) = [x'_i(n), x'_i(n-1), \dots, x'_i(n-L+1)]^T. \quad (2)$$

The weight vector $W(n)$ of the filter is defined as:

$$W(n) = [W_1^T(n), W_2^T(n), \dots, W_J^T(n)]^T; \quad (3)$$

where the weight coefficient of the j -th filter $W_j(n)$ is:

$$W_j(n) = [w_{j,1}(n), w_{j,2}(n), \dots, w_{j,L}(n)]^T. \quad (4)$$

Defining the secondary acoustic signal output by the controller as:

$$r(n) = [r_1(n), r_2(n), \dots, r_J(n)]. \quad (5)$$

The output signal $r_j(n)$ of the j -th secondary sound source at the n -th moment is given by:

$$r_j(n) = \sum_l^L w_{j,l}(n) \times x(n-l+1) = w_j^T \times x(n). \quad (6)$$

To sum up, the output $r(n)$ of the filter can be obtained as follows:

$$r(n) = X^T(n)W(n); \quad (7)$$

where, $X(n)$ is a matrix given as follows:

$$X(n) = \begin{bmatrix} x(n) & 0 & \dots & 0 \\ 0 & x(n) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & x(n) \end{bmatrix}. \quad (8)$$

The transfer function estimated value $H'(n)$ of the secondary channel expressed as:

$$H'(n) = \begin{bmatrix} h_{11}(n) & h_{12}(n) & \dots & h_{1K}(n) \\ h_{21}(n) & h_{22}(n) & \dots & h_{2K}(n) \\ \vdots & \vdots & \ddots & \vdots \\ h_{J1}(n) & h_{J2}(n) & \dots & h_{JK}(n) \end{bmatrix}. \quad (9)$$

A sub-element $h_{jk}(n)$ can be regarded as the weight vector of the filter with length L , i.e.,

$$h_{jk}(n) = [w_{jk1}(n), w_{jk2}(n), \dots, w_{jkL}(n)]^T. \quad (10)$$

Defining $y(n)$ as:

$$y(n) = [y_1(n), y_2(n), \dots, y_K(n)]^T. \quad (11)$$

The k -th element of $y(n)$, denoted as $y_k(n)$, is the superposition of acoustic signals from J loudspeakers to the k -th error sensor through the secondary channel, which is expressed in the form of convolution as follows:

$$y_k(n) = \sum_{j=1}^J h_{jk}(n) \times r_j(n). \quad (12)$$

Subsequently, the input vector of secondary acoustic signal to error sensor $y(n)$ can be written as:

$$y(n) = H^T(n) \times r(n). \quad (13)$$

The multi-channel system can cancel the primary noise signal by minimizing the sum of squares of K error signals. The weight vector composed of the output of K error sensors is marked as $e(n)$, shown in the next equation.

$$e(n) = d(n) + y(n); \quad (14)$$

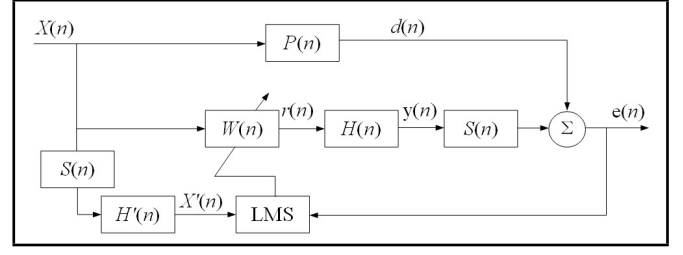


Figure 2. Block diagram of the FxLMS algorithm with matrix decoupling.

where, $d(n)$ is the expected signal to be approximated by the filter. The weight vector iteration formula of the multi-channel FxLMS algorithm can be obtained as follows:

$$W(n+1) = W(n) - 2\eta_1 X'(n)e(n); \quad (15)$$

where, η_1 is the learning rate, and $X'(n)$ is the filtered reference signal matrix. If it is simplified into a scalar form, the iterative formula for the l -th weight coefficient $w_{jil}(n)$ of the i -th reference signal and the j -th loudspeaker in the controller is given as:

$$w_{jil}(n+1) = w_{jil}(n) - 2\mu \sum_{k=1}^K y_k(n)x'_{ki}(n-l+1); \quad (16)$$

$$x'_{ki}(n) = \sum_{l=1}^L w_{jkl}(n)x_i(n-l+1). \quad (17)$$

The mutual coupling among channels is considered in the multi-channel FxLMS algorithm and reflected in the transfer function of secondary channels. However, the stability of the system and its control performance are affected by this coupling, therefore, it is necessary to decouple the control system.

2.2. FxLMS Algorithm with Matrix Decoupling

The problem of eliminating channels coupling is transformed into constructing a decoupling matrix. Based on the principle of feed-forward compensation,¹⁵ a decoupling matrix is constructed to render the off-diagonal element of the transfer function matrix of the secondary channel equal to 0, resulting in complete decoupling. The decoupled system is equivalent to multiple single loops, and the output and error signals have one-to-one correspondence. Figure 2. presents the block diagram of the FxLMS algorithm with matrix decoupling.

The diagonal elements of the decoupling matrix $S(n)$ constructed in this paper are all equal to one in order to reduce the computational complexity of the decoupling matrix, which enables it to meet the following requirements:

$$S(n)H'^T(n) = \hat{H}(n); \quad (18)$$

where, $H(n)$ is the transfer function of secondary channel, and $\hat{H}(n)$ is a diagonal matrix. The expression of Eqn. (2.18) can

be written as:

$$S(n)H'^T(n) = \begin{bmatrix} 1 & S_{21}(n) & S_{31}(n) \\ S_{12}(n) & 1 & S_{32}(n) \\ S_{13}(n) & S_{23}(n) & 1 \end{bmatrix}.$$

$$\begin{bmatrix} h_{11}(n) & h_{21}(n) & h_{31}(n) \\ h_{12}(n) & h_{22}(n) & h_{32}(n) \\ h_{13}(n) & h_{23}(n) & h_{33}(n) \end{bmatrix} = \begin{bmatrix} \hat{H}_1(n) & 0 & 0 \\ 0 & \hat{H}_2(n) & 0 \\ 0 & 0 & \hat{H}_3(n) \end{bmatrix} = \hat{H}(n); \quad (19)$$

The decoupling matrix is equivalent to superimposing the output of a single controller onto the other output signals in order to offset their influence. The new transfer function matrix of the error channel is transformed into a diagonal matrix, by calculation, the decoupling matrix $S(n)$ is

$$S(n) = \begin{bmatrix} 1 & \frac{h_{23}(n)h_{31}(n)-h_{21}(n)h_{33}(n)}{h_{22}(n)h_{33}(n)-h_{23}(n)h_{32}(n)} & \frac{h_{13}(n)h_{31}(n)-h_{11}(n)h_{33}(n)}{h_{12}(n)h_{33}(n)-h_{13}(n)h_{32}(n)} \\ \frac{h_{13}(n)h_{32}(n)-h_{12}(n)h_{33}(n)}{h_{11}(n)h_{33}(n)-h_{13}(n)h_{32}(n)} & 1 & \frac{h_{13}(n)h_{21}(n)-h_{11}(n)h_{23}(n)}{h_{11}(n)h_{22}(n)-h_{12}(n)h_{21}(n)} \\ \frac{h_{12}(n)h_{31}(n)-h_{11}(n)h_{32}(n)}{h_{11}(n)h_{22}(n)-h_{12}(n)h_{21}(n)} & \frac{h_{21}(n)h_{32}(n)-h_{22}(n)h_{31}(n)}{h_{22}(n)h_{33}(n)-h_{23}(n)h_{32}(n)} & 1 \end{bmatrix}; \quad (20)$$

Using the expression of $S(n)$, a new secondary channel matrix $\hat{H}(n)$ can be expressed as follows:

$$\hat{H}(n) = \begin{bmatrix} h_{11}(n) + h_{12}(n) \times S_{21}(n) + h_{13}(n) \times S_{31}(n) & 0 & 0 \\ 0 & h_{22}(n) + h_{21}(n) \times S_{12}(n) + h_{23}(n) \times S_{32}(n) & 0 \\ 0 & 0 & h_{33}(n) + h_{31}(n) \times S_{13}(n) + h_{32}(n) \times S_{23}(n) \end{bmatrix}; \quad (21)$$

By inserting the decoupling matrix before the transfer function matrix of the secondary channel, the system can become roughly equivalent to multiple single loops. The weight coefficient of a certain controller is updated only based on the corresponding error signal, then the computational complexity of the control system is reduced, and the stability of the algorithm is improved.

However, the matrix decoupling method is only suitable for a small number of channels. The calculation of the decoupling matrix may be more complicated for many channels. Moreover, as a traditional decoupling method, matrix decoupling relies heavily on the precise mathematical model of the controlled object. Many actual noise control systems are nonlinear and time-varying complex multivariable systems, and as a result, traditional decoupling methods are not suitable for achieving adaptive decoupling. Therefore, a better decoupling method is needed to eliminate the channel coupling problem.

2.3. FxLMS Algorithm Combined with the PID Neural Network

The multi-channel FxLMS algorithm is combined with the intelligent decoupling method, i.e., the PID neural network de-

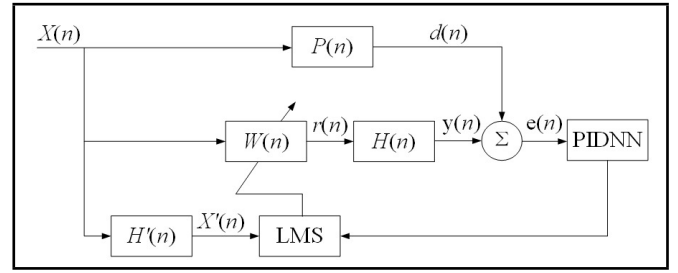


Figure 3. Block diagram of the FxLMS algorithm combined with the PID neural network.

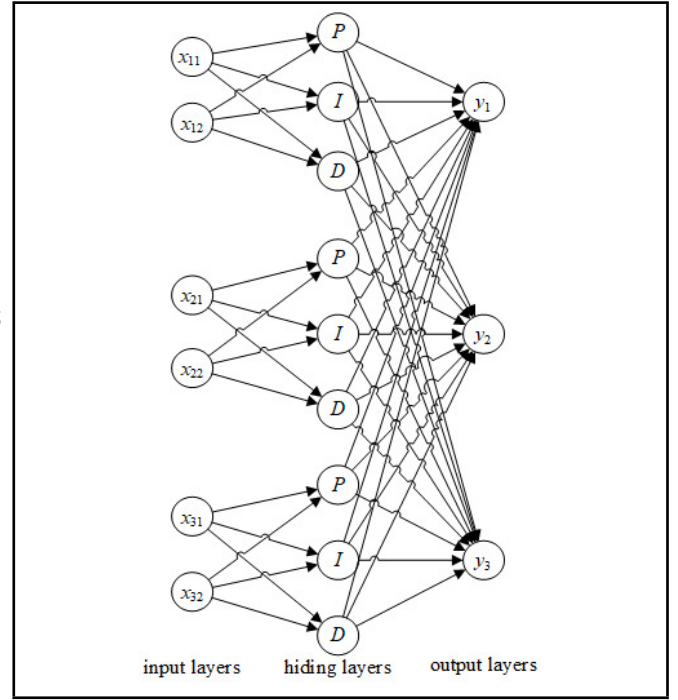


Figure 4. Structural chart of the PID neural network.

coupling, to solve the coupling problem among channels. Figure 3. shows the block diagram of the FxLMS algorithm combined with the PID neural network decoupling method.

According to Fig. 4, the PID neural network can be divided into three layers: input layer, hidden layer and output layer. The PID neural network with m control variables contains m parallel identical sub-networks. The sub-networks are not only independent of one another, but also interconnected via network connection weights.

Figure 4 depicts that there are two neurons in the input layer of each sub-network, which receive the target and current values of the control quantity. The variables x_{11} , x_{21} and x_{31} , and x_{12} , x_{22} and x_{32} are the control targets and current values of the control quantity. The hidden layer of each sub-network is composed of a proportional element P , an integral element I and a differential element D , corresponding to proportional control, integral control and differential control in a PID controller, respectively. The output of each neuron in the hidden layer is given by u . The output layer of each sub-network has only one neuron, namely y_1 , y_2 and y_3 are presented in Fig. 4, which represents the control law calculated by the multi-layers neural network. Furthermore, s is the serial number of the parallel sub-network, i is the serial number of the input layer neurons in the sub-network, j is the serial number of the hidden layer

Table 1. Optimal weights of proportional, differential and integral elements.

Parameter	Channel 1			Channel 2			Channel 3		
Number of hidden layer channels	1	2	3	1	2	3	1	2	3
Weight of proportional element	0.0225	0.0232	0.0067	0.0292	0.0072	0.0064	0.0238	0.0024	0.0095
Weight of differential element	0.0214	0.0012	0.0019	0.0288	0.0191	0.0028	0.0070	0.0244	0.0113
Weight of integral element	0.0076	0.0269	0.0278	0.0219	0.0216	0.0219	0.0299	0.0228	0.0153

Table 2. Amplitude of noise signal before and after the use of the three algorithms.

Algorithm	Amplitude of input signal	Amplitude of error signal	Ratio of noise reduction	Number of convergent iterations
Traditional FxLMS	2.1349	0.0576	97.3%	1870
Matrix decoupling	2.1349	0.0450	97.8%	1864
PID neural network decoupling control	2.1349	0.0046	99.7%	750

neurons in the sub-network, and k is the serial number of the output layer neurons in the sub-network. As for multi-layers neural network, ω_{ij} is the weight between the input and hidden layers, and ω_{jk} is the weight between the hidden and output layers.

Then, the input layer and hidden layer inputs are as follows:

$$x_{si}(n) = e(n); \quad (22)$$

$$net_{sj}(n) = \sum_{i=1}^2 \omega_{ij} x_{si}(n); \quad j = 1, 2, 3; \quad (23)$$

and the outputs of the proportional neurons, integral neurons and differential neurons are given as:

$$u_{s1}(n) = net_{s1}(n); \quad (24)$$

$$u_{s2}(n) = net_{s2}(n) + u_{s2}(n-1); \quad (25)$$

$$u_{s3}(n) = net_{s3}(n) - net_{s3}(n-1). \quad (26)$$

The result generated by the output layer can be represented by the following expression:

$$y_h(n) = \sum_{k=1}^3 \sum_{j=1}^3 \omega_{jk} u_{sj}(n). \quad (27)$$

The error update formula can be written as:

$$J = \sum E = \sum_{n=1}^k [y_h(n) - p(n)]^2. \quad (28)$$

The weights between the output layer to the hidden layer, and the input layer to the output layer are updated as follows:

$$\omega_{jk}(n+1) = \omega_{jk}(n) - \eta_2 \frac{\partial J}{\partial \omega_{jk}}; \quad (29)$$

$$\omega_{ij}(n+1) = \omega_{ij}(n) - \eta_2 \frac{\partial J}{\partial \omega_{ij}}; \quad (30)$$

where, η_2 is the learning rate.

It is necessary to adjust the relationship among the proportional element, integral element, and differential element for improving the performance of the PID controller. However, this is not a simple linear relationship, but a search for the best relationship in a constantly changing nonlinear combination.

The neural network can approximate any nonlinear function, and its structure and learning algorithm are simple and clear. It can find the optimal PID control parameters through the network.

The calculation process of the PID neural network can be divided into two processes: the forward calculation and the backward error transmission of the network and weight correction. This process is repeated until the error is eliminated. As long as the input signals are present, they are passed through the network. When passing through a layer, the input signals are used to calculate the corresponding weight based on the weight of the current node and the activity function. The output results are transferred to the next layer. When the current network output is inconsistent with the expected output, the error signals between the output and expected output are transmitted in the reverse direction and the error loop is returned as the basis for modifying the weights. The control system can realize the smallest mean square error according to the weight control output signals.

3. RESULTS AND ANALYSIS

In this section, the convergence performance of the proposed multi-channel active control algorithm is verified by comparing the amplitude of the error signal after control and the convergence rate of the algorithm during the control process. The simulations are carried out using Matlab. By simulation analysis, the decoupling control of the FxLMS algorithm combined with the PID neural network is compared with the traditional FxLMS algorithm and the FxLMS algorithm with matrix decoupling. In the simulation, the filter length is set as 16, the number of iterations is set to 5000, the step is 0.0001, the learning rate η_1 is 0.0001, and the learning rate η_2 is 0.001. It can be observed in Fig. 5(a) that the input signal is nonstationary noise, that the sum of sinusoidal signals of amplitude is equal to 2, and the input signal can be written as:

$$x(n) = \sin(0.24\pi t + 0.2\pi) + \sin(0.34\pi t) + \sin(0.5\pi t); \quad (31)$$

where, t is time.

The control results of the three algorithms are presented in Figures 5., 6. and 8. The abscissa is the number of iterations, and the ordinate is the amplitude of noise. Moreover, Figure 9. compares the residual error signals obtained from the three

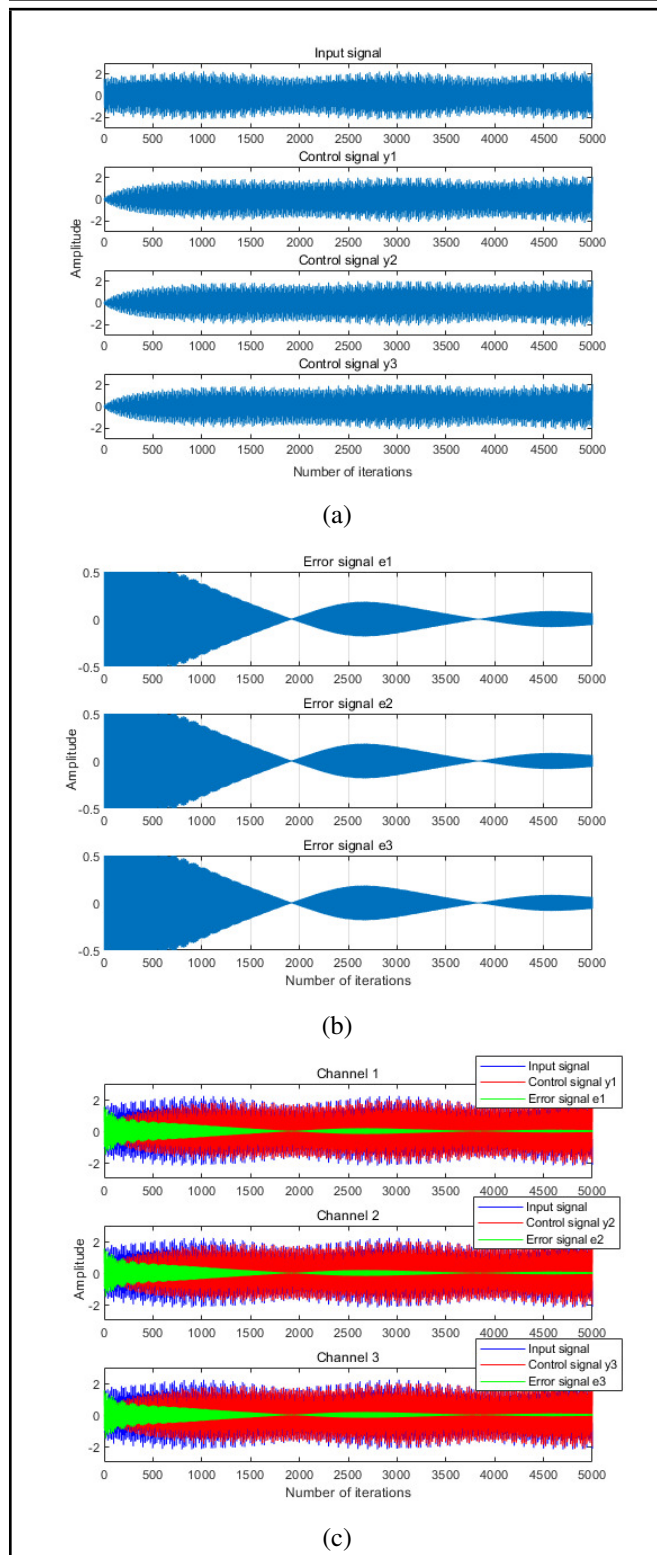


Figure 5. Traditional FxLMS algorithm, (a) input and control signals, (b) error signals, and (c) control results of three channels.

algorithms. The amplitude of the noise signal before and after the control results with the three algorithms is listed in Table 1.

Figure 5. depicts the control results of the traditional FxLMS algorithm. The control system converges when the number of iterations is about 2000. After convergence, the amplitude of the error signal is 0.0576.

According to Figure 6., the FxLMS algorithm with matrix decoupling converges when the number of iterations is about

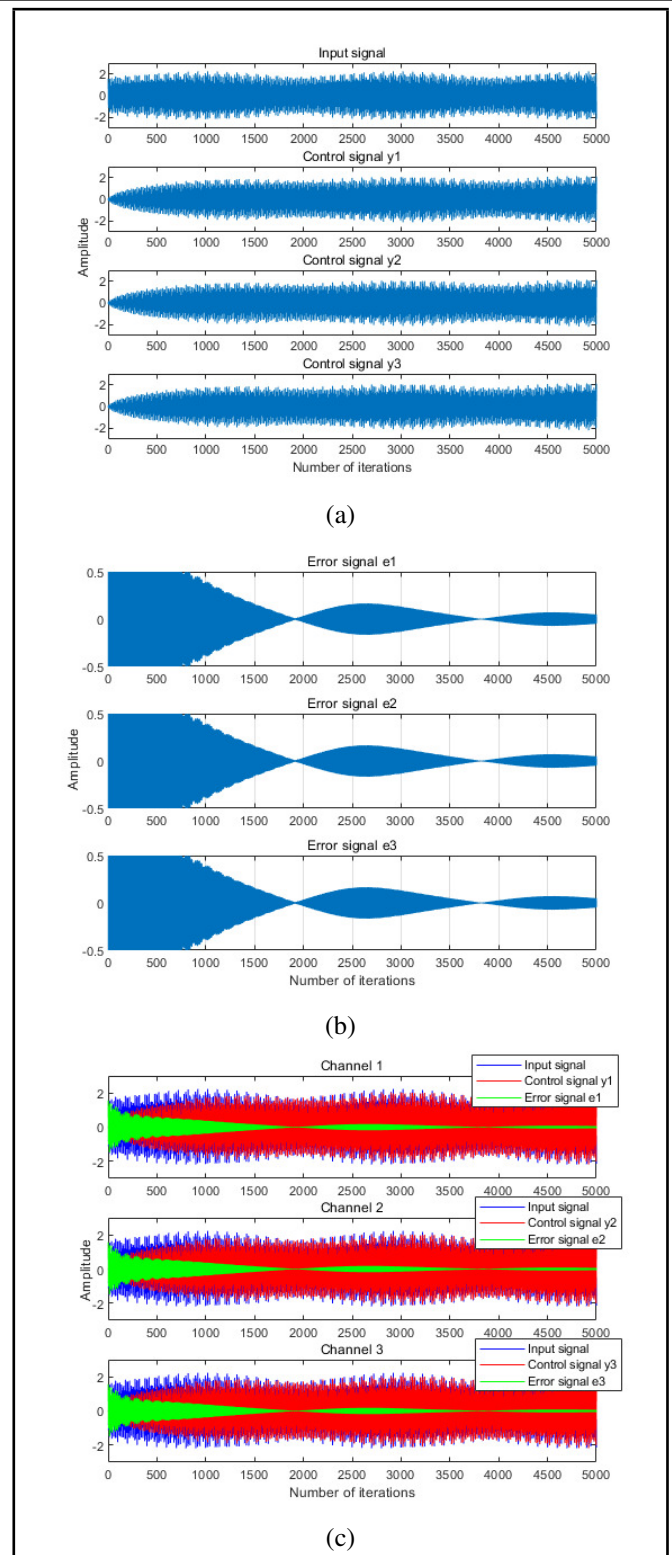


Figure 6. FxLMS algorithm with matrix decoupling, (a) input and control signals, (b) error signals, and (c) control results of three channels.

2000. After convergence, the error signal amplitude is equal to 0.0450.

In the proposed algorithm combining the FxLMS algorithm with the PID neural network, the initial weights of the proportion, integral and differential units are random numbers. The optimal weights that make the convergence reach the optimal control are presented in Figure. 7., and the specific values are taken after four decimal places, as shown in Table 1.

Figure 8 shows that the FxLMS algorithm combined with

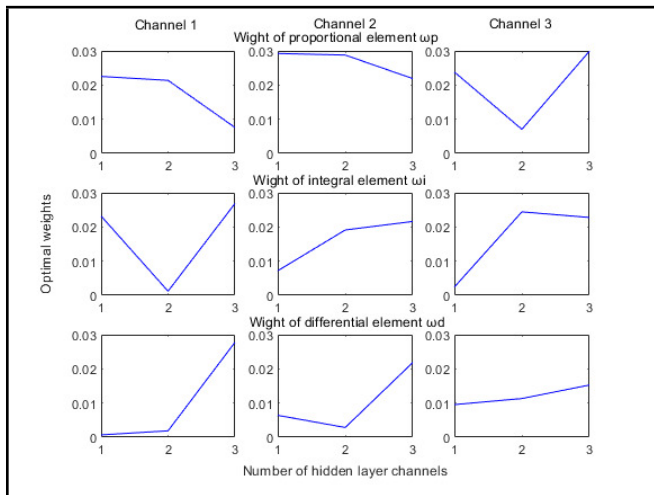


Figure 7. Optimal weights of proportional, differential and integral elements.

the PID neural network converges when the number of iterations is about 800, and the residual amplitude is 0.0046.

Figure 9. compares the residual error signals for the three algorithms. Figure 9(b). gives an enlarged view of the residual error results. It can be observed from the figure that the FxLMS algorithm combined with the PID neural network has the best control performance, followed by the FxLMS algorithm with matrix decoupling and the traditional FxLMS algorithm.

Table 2 summarizes the amplitude of the initial signal and the residual error after convergence of the three algorithms and number of convergent iterations.

The simulation results show that the FxLMS algorithm combined with the PID neural network decoupling control has better control performance and convergence rate for the active control of multi-channel noise. The traditional FxLMS algorithm converges at about 1870 iterations, while the FxLMS algorithm with matrix decoupling converges at about 1865 iterations, while the proposed algorithm converges when the number of iterations is about 750. Thus, the proposed algorithm has a faster convergence rate.

According to Figure 9. and Table 2, the input signals of the three algorithms have the same amplitude of 2.1349. The error signal amplitude of the traditional FxLMS algorithm is 0.0576, and the noise reduction is about 97.3%. The error signal amplitude of the FxLMS algorithm with matrix decoupling is 0.0450, and the noise reduction is about 97.8%. The residual amplitude of the FxLMS algorithm combined with the PID neural network is 0.0046, and the noise reduction is about 99.7%. The results show that the control performance of the FxLMS algorithm with matrix decoupling was slightly better than the traditional algorithm FxLMS algorithm, while the performance of the proposed algorithm was better than the other two algorithms.

Currently, the ANC algorithms of the FxLMS are based on the linear control theory. However, the ANC system often shows nonlinear characteristics. A neural network has strong self-adaptation and self-learning abilities, and can approximate nonlinear functions with arbitrary precision, as well as deal with nonlinear problems. Therefore, the FxLMS algorithm combined with the neural network algorithm can achieve a better control performance. In addition, the three control units in the PID controller complement one another, and the comple-

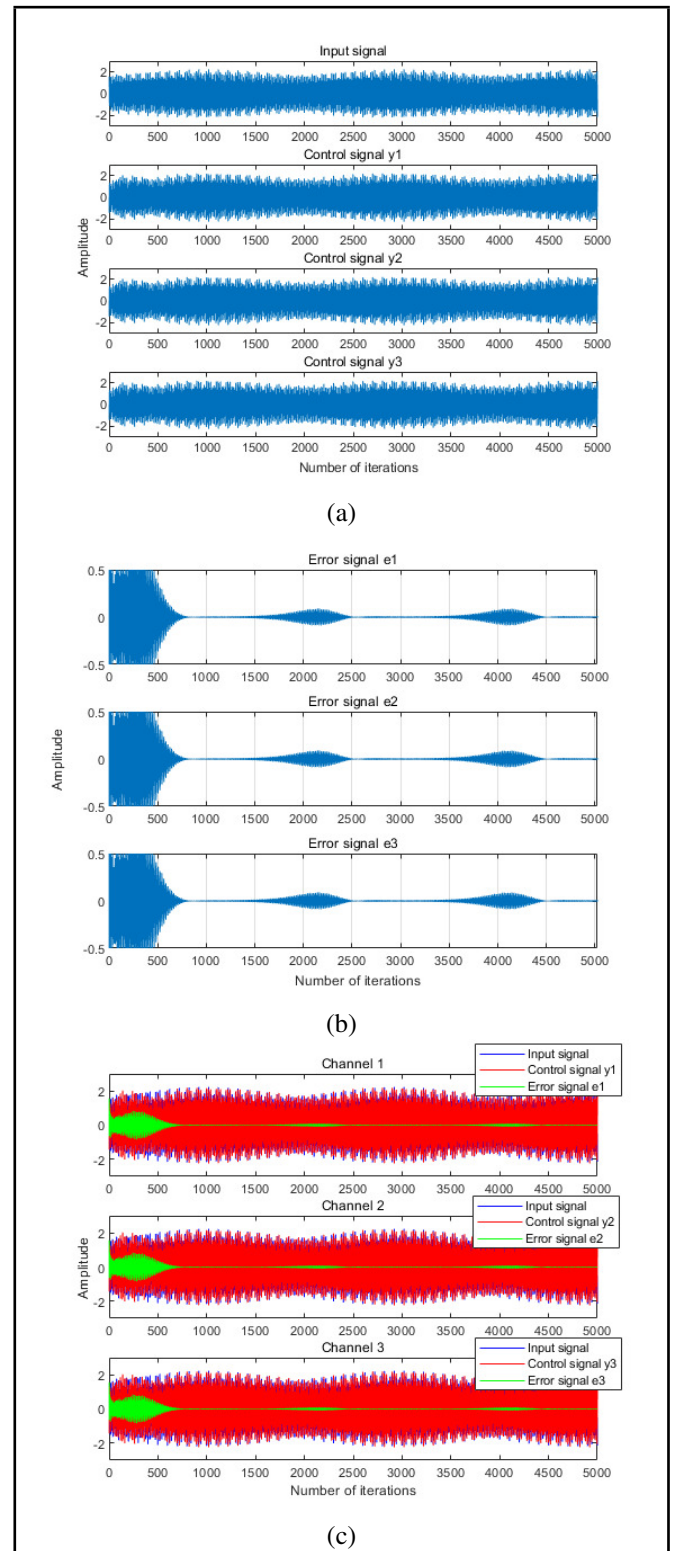


Figure 8. FxLMS algorithm combined with the PID neural network, (a) input and control signals of three channels, (b) error signals of three channels, (c) control results of three channels.

ment one another based on proportional control. The integral control can eliminate the steady-state error but may increase the overshoot. Differential control can accelerate the response speed of the system and weaken the overshoot trend. Therefore, the FxLMS algorithm combined with the PID neural network has better control performance and faster convergence speed.

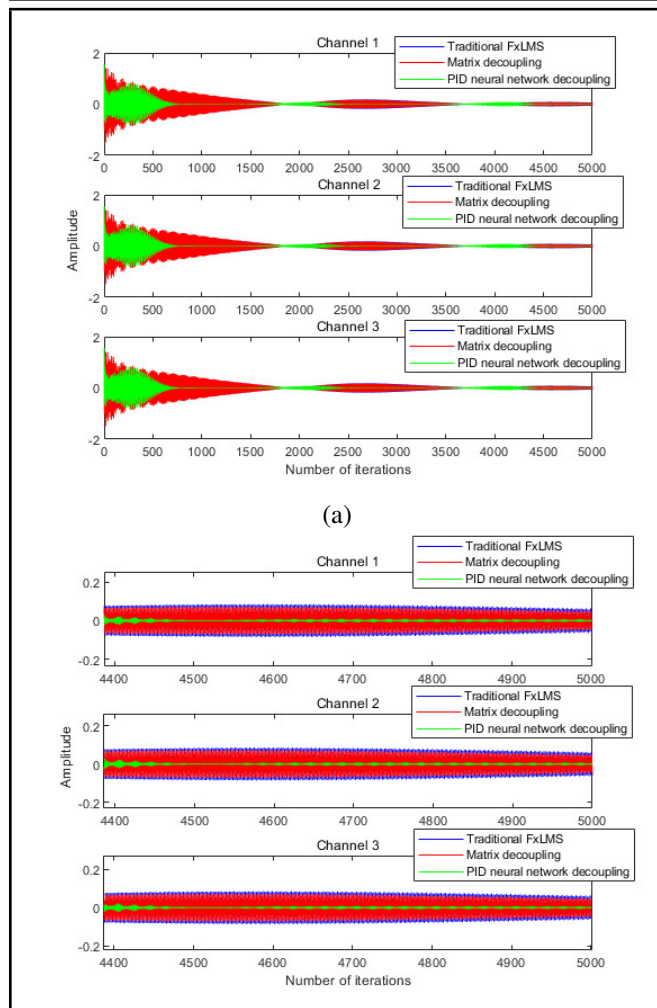


Figure 9. Comparison of control results for the three algorithms, (a) residual error signals for the three algorithms, (b) the enlarged view of the residual error results.

4. CONCLUSIONS

By applying the PID neural network algorithm to the multi-channel ANC system, this paper proposed a novel algorithm. As the number of channels increased in the multi-channel ANC system, the stability of the control system was affected by the presence of multi-channel coupling. Thanks to the unique non-linear mapping structure of the neural network, almost any nonlinear function can be approximated and the mutual interference among channels can be eliminated. In addition, the neural network takes a very short time to calculate these coefficients because of its parallel structure. It can be combined with PID control to improve the convergence speed of the control system. This algorithm combines control and decoupling as a problem to be solved together. Simulation results and comparison analysis of results from the traditional FxLMS and the matrix decoupling FxLMS showed that the proposed algorithm had better control performance and faster convergence speed. In the follow-up real vehicle test research, it is necessary to model the secondary path first. The fuzzy control method proposed by Kurczyk²² also provides a new idea. Consideration can be given to combine the fuzzy control with the algorithm proposed in this paper to omit the process of secondary path modeling. Moreover, the noise signal targeted in the simulation in this paper was relatively simple, and the control of

the complex signal has not been studied yet. Another research area in the multi-channel ANC system is the increase in the complexity of the control system as the number of channels increases. Further research will be carried out to demonstrate control over more complex interior noise signals and a study to reduce the computational complexity of the control algorithm.

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